

LIGHTKEY

Professional DMX Lighting Control Software

Exclusively Designed for macOS



Prototyping Demo Project

Nuzm Solutions — Prototyping Group

 **Platform** macOS
Exclusive

 **Protocol** DMX512
Industry Std

 **Category** Lighting
Control

 **Used By** Stage &
Event Pros

What is Lightkey?

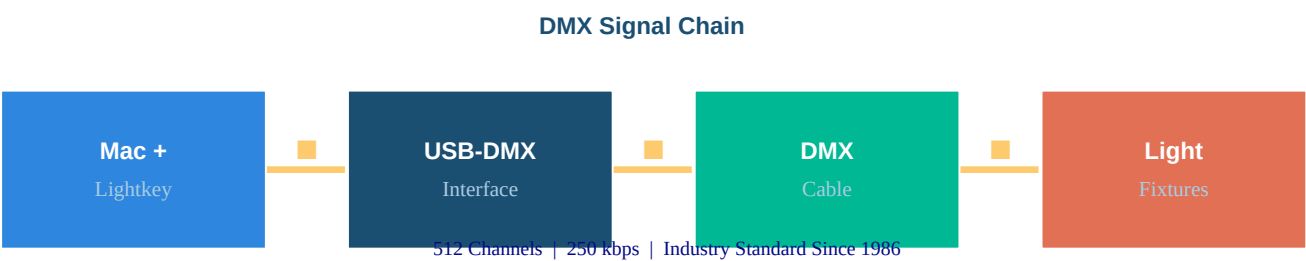
Lightkey is a powerful, macOS-native DMX lighting control application used by professionals in stage, theater, concerts, and architectural lighting. It provides real-time control of DMX fixtures with an intuitive interface, timeline editor, and a massive fixture library — all optimized for Apple hardware.

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1. What is DMX Protocol?

DMX512 (Digital Multiplex) is the universal communication standard used to control stage lighting and special effects equipment. Originally developed in 1986, it allows a single controller to manage up to 512 individual channels — each controlling one parameter of a light fixture (brightness, color, focus, etc.).



DMX Channel Breakdown (Example: RGB Moving Head)

Channel	Function	Value Range	Effect
CH 1	Dimmer / Intensity	0 – 255	Controls brightness
CH 2	Red	0 – 255	Red color intensity
CH 3	Green	0 – 255	Green color intensity
CH 4	Blue	0 – 255	Blue color intensity
CH 5	Pan (Horizontal)	0 – 255	Left/Right movement
CH 6	Tilt (Vertical)	0 – 255	Up/Down movement
CH 7	Strobe	0 – 255	Flash speed control

2. Lightkey Interface Overview

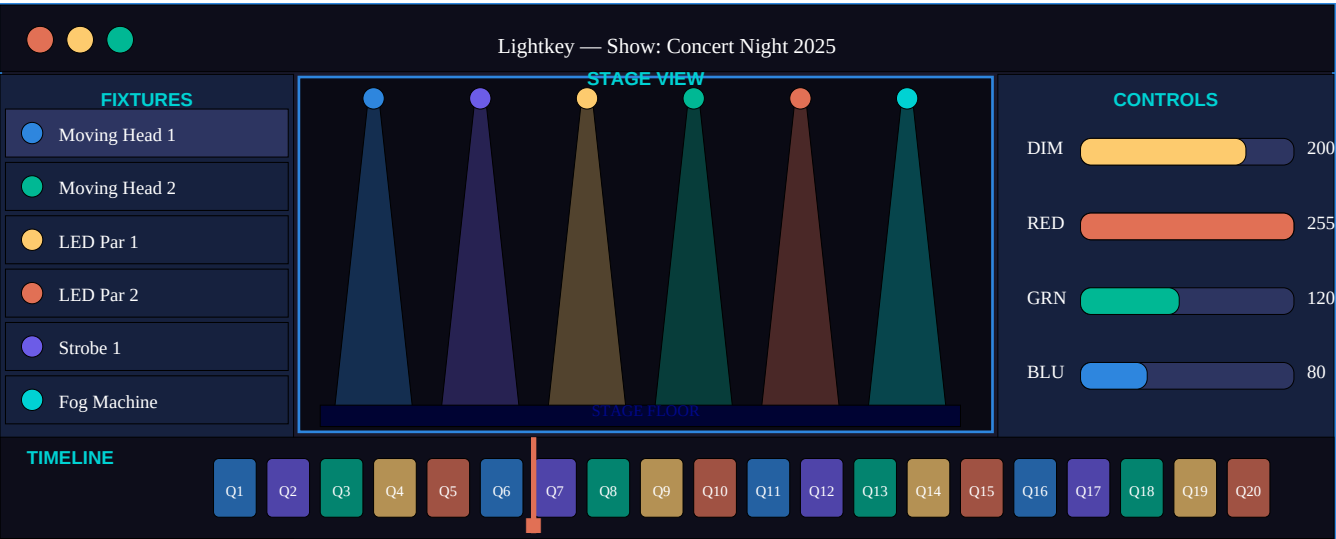


Fig 1: Lightkey Interface — Fixture Panel | Stage View | Controls | Timeline

3 System Architecture

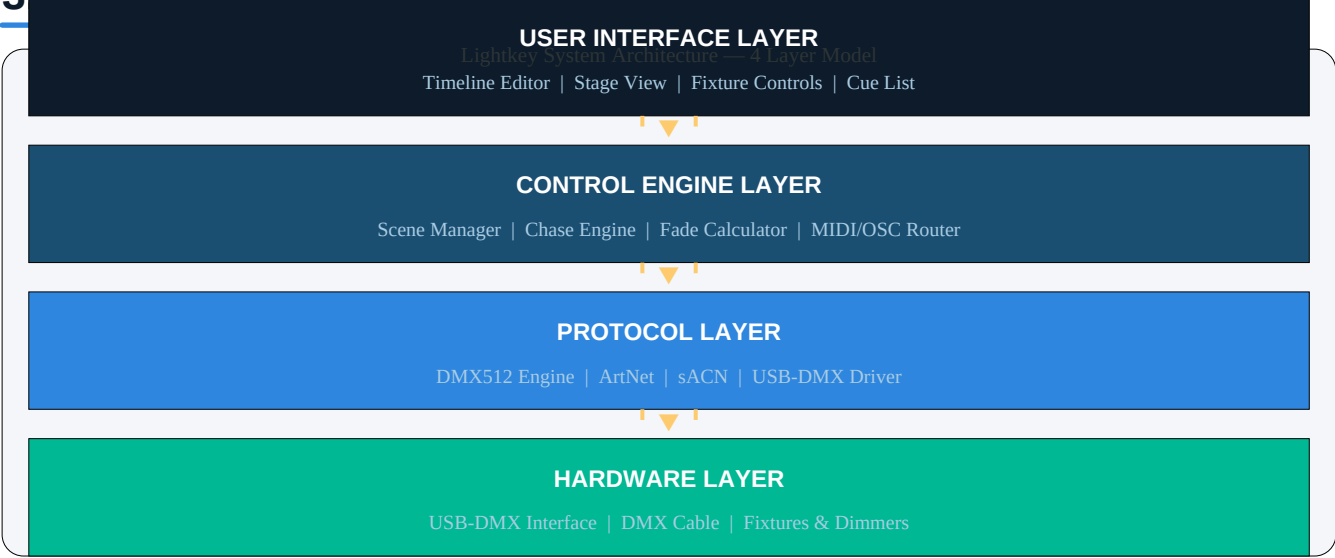


Fig 2: Lightkey 4-Layer System Architecture

4. Key Features (Detailed)

	Full DMX512 Control Controls all 512 channels of a DMX universe with real-time, zero-latency response. Supports multiple universes via ArtNet and sACN protocols for large installations.
	Massive Fixture Library Ships with 10,000+ pre-programmed fixtures from brands like Chauvet, Martin, Elation, Robe, and Clay Paky. Import custom fixture profiles in GDTF or generic DMX format.
	Timeline & Cue Editor Visual drag-and-drop timeline lets you program complex multi-scene light shows with precise timing. Supports smooth fade curves, hold times, and looping sequences.
	Live Performance Mode Dedicated live mode with instant scene triggering, flash buttons, speed controls, and BPM sync. Perfect for DJs, bands, and theater operators.
	MIDI & OSC Integration Map any parameter to MIDI controllers, keyboards, or OSC-enabled apps. Use a MIDI fader board to control intensities, or OSC from an iPad to trigger cues wirelessly.
	Native macOS Optimization Built with SwiftUI for macOS — silky smooth 60fps interface, Apple Silicon support (M1/M2/M3), Dark Mode, and full compatibility with macOS Sonoma and Ventura.

5. Step-by-Step Workflow

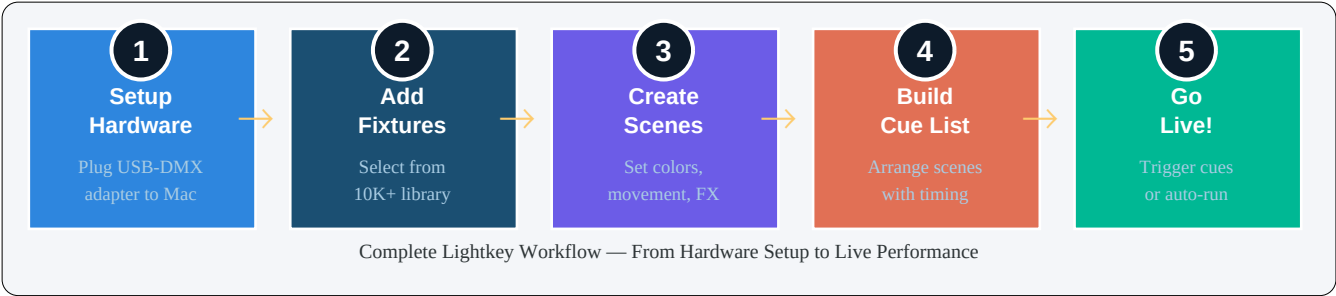


Fig 3: Lightkey 5-Step Workflow

6. Live Demo Scenario — Concert Setup

The following scenario demonstrates how Lightkey would be used for a live concert with 3 moving heads, 4 LED PAR cans, and 1 strobe light — a typical small venue setup.

#	Cue Name	Fixtures Active	Color	Effect	Trigger
1	House Open	All LED PARs	Warm White	Static, 80%	Manual
2	Band Intro	Moving Heads	Deep Blue	Slow sweep	Manual
3	Song Start	All Fixtures	Red + White	Beat chase	BPM Sync
4	Chorus Hit	All + Strobe	Full RGB	Fast strobe	Manual
5	Solo Spot	MH Center	Amber	Follow spot	Manual
6	Finale	All Fixtures	Rainbow	Full chase	Auto 8s
7	Blackout	None	—	Instant off	Manual

Table: Concert Cue List — 7 Cues for a Live Show

7. Competitor Comparison

Feature	Lightkey (macOS)	QLC+ (Cross-platform)	DMXControl (Windows)	GrandMA3 (Console)
Platform	macOS Only	Win/Mac/Linux	Windows Only	Hardware
Price	Free + Pro \$149	Free	Free	\$5,000+
Ease of Use	★★★★	★★	★★	★★
Fixture Library	10,000+	5,000+	3,000+	15,000+
Timeline Editor	■ Yes	■ Yes	■ Yes	■ Yes
MIDI Support	■ Yes	■ Yes	■ Yes	■ Yes
Apple Silicon	■ Native	■■ Rosetta	■ No	■ No
Best For	Mac Users Pros & Events	Budget Users Hobbyists	Windows Only Basic Use	Large Venues Big Budget

8. Use Cases

■

Theater & Stage

■

Live Concerts & Events

■■

Architectural Lighting

■

TV & Film Studios

■

Houses of Worship

■

DJ & Club Lighting

9. Pros & Cons

■ PROS	■ CONS
■ Native macOS app — smooth & stable performance	■ macOS only — no Windows or Linux support
■ Intuitive drag-and-drop interface, easy to learn	■ Advanced features locked behind paid Pro license
■ Massive fixture library with 10,000+ profiles	■ Single DMX universe in free version
■ Free version available with core features	■ No built-in fixture editor (need external tool)
■ Full Apple Silicon (M1/M2/M3) support	■ Requires USB-DMX interface (additional hardware cost)
■ Strong MIDI/OSC integration for hardware control	■ Not ideal for very large professional venues
■ Regular updates and active developer support	
■ ArtNet & sACN for advanced network DMX	

10. Conclusion

Lightkey stands out as the **premier DMX lighting control solution for macOS users**. Whether you are a hobbyist setting up your first light show or a professional lighting designer running a complex stage production, Lightkey delivers the tools you need in an intuitive, native macOS package.

Its combination of a massive fixture library, powerful timeline editor, real-time performance capabilities, and seamless MIDI/OSC integration makes it the go-to choice for Mac-based lighting professionals worldwide. The free tier makes it accessible for beginners, while the Pro license unlocks its full potential for serious productions.

Verdict: If you own a Mac and need professional DMX control — Lightkey is the answer. ■